Task-Aware Adaptation and Validation-Time Oracle Analysis for Text-Rich Multimodal Reasoning

Hongrong Yang
School of Science and Engineering
The Chinese University of Hong Kong, Shenzhen
Shenzhen, China
225010041@link.cuhk.edu.cn

Che Cheng
School of Science and Engineering
The Chinese University of Hong Kong, Shenzhen
Shenzhen, China
225015050@link.cuhk.edu.cn

Shaoming Li
School of Science and Engineering
The Chinese University of Hong Kong, Shenzhen
Shenzhen, China
224010203@link.cuhk.edu.cn

Abstract

Text-rich multimodal reasoning requires models to read visual text, reason over structured layouts, and produce answers in task-specific formats. This work presents a unified framework for this setting based on Qwen3-VL-8B-Instruct. The framework standardizes heterogeneous datasets into a shared sample format and uses structured prompts to improve input-output consistency. It then follows a progressive adaptation pipeline that combines PEFT-based supervised tuning, a task-aware Text-Rich Adapter, and task-stratified GRPO. We also report reward-guided Best-of-N as a validation-time oracle analysis: candidate answers are selected with gold-reference task rewards, so the result is an upper bound rather than a deployable verifier. Experiments on in-domain and external benchmarks show that the proposed framework improves target-task performance while preserving competitive external generalization. The final GRPO checkpoint achieves the best in-domain overall score of 0.8535. Error analysis shows clear reductions in text reading and chart reasoning failures. Code is available at `https://github.com/however-yang/TextRich-Reasoning-MLLM`.

1 Introduction

Text-rich multimodal reasoning asks a model to answer questions about images with dense text and structured visual content. Given an image I and a question q, the model needs to generate an answer y. The image may be a document, a chart, a table, or a science figure. This task is harder than general image question answering. The model must read visual text, understand layout, and produce answers in the required format. For example, DocVQA and ChartQA often need short text or numeric answers, while ScienceQA and MMMU use multiple-choice answers [1–4]. This setting therefore requires both visual understanding and strict answer control.

Large multimodal models provide a useful starting point, but direct use is often not enough for this setting. Text-rich tasks have different data formats, answer types, and evaluation metrics. A single training stage may not handle these differences well. Full fine-tuning is also costly for an 8B-scale backbone. To address these issues, this work builds a unified framework based on Qwen3-VL-8B-Instruct. The framework first maps different datasets into a shared sample format. It then uses structured prompts to make the input and output format more stable.

The framework follows a progressive adaptation path. It first compares LoRA and DoRA for supervised adaptation [5, 6]. It then adds a lightweight Text-Rich Adapter to selected decoder layers. This adapter uses task embeddings, a bottleneck transformation, and a gated residual path. After supervised training, the framework applies task-stratified GRPO. Each reward group comes from one task, so reward comparison stays within the same metric type. Finally, reward-guided Best-of-N is used only as validation-time oracle analysis, because candidate selection uses reference answers.

This paper makes the following contributions:

- We build a unified framework for text-rich multimodal reasoning. It includes data formatting, structured prompting, training, and inference.

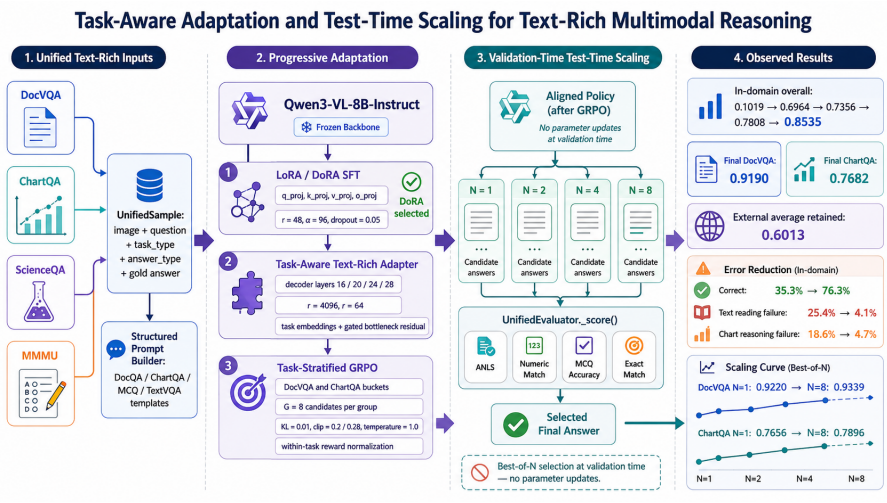

Figure 1: Overview of the proposed training and inference pipeline for text-rich multimodal reasoning.

- We design a progressive adaptation path for Qwen3-VL-8B-Instruct. It combines LoRA or DoRA, a Text-Rich Adapter, and task-stratified GRPO.
- We study reward-guided Best-of-N as validation-time oracle analysis. It generates multiple answers and selects the one with the highest reference-based task reward on validation data.

2 Related Work

Text-rich multimodal reasoning focuses on images with dense text and structured visual content. This setting includes documents, charts, tables, science figures, and diagrams. Several benchmarks define this problem from different views, including document QA, chart QA, science QA, and expert-level multimodal QA [1–4]. Earlier methods show the need for structure-aware modeling. LayoutLMv3 studies document pre-training with text and image masking [7]. Donut maps document images to output sequences without external OCR [8]. DePlot converts charts into tables before language-based reasoning [9]. These works show that text-rich inputs need visual reading and layout understanding. They also motivate the use of a general multimodal backbone for this setting.

Model adaptation is closely related to our framework. Large models are costly to fully fine-tune, so many studies update only a small part of the model. Adapter tuning inserts small trainable modules into pre-trained models [10]. Prompt tuning learns continuous prompts while keeping the model mostly frozen [11]. LoRA freezes the original weights and learns low-rank updates [5]. QLoRA further reduces memory cost through quantized low-rank adaptation [12]. DoRA improves LoRA by separating weight magnitude and direction [6]. Our framework follows this direction but uses it for text-rich multimodal reasoning. It first compares LoRA and DoRA in supervised adaptation. It then inserts a lightweight Text-Rich Adapter into selected decoder layers. This adapter uses task embeddings, a bottleneck transformation, and a gated residual path. This design follows the actual project implementation. It supports different text-rich tasks with one shared backbone.

Reward-based training provides another way to improve model behavior after supervised learning. PPO is a common policy optimization method with a clipped objective [13]. RLHF uses human preference data to align language models with desired outputs [14]. DPO offers a simpler preference optimization method without explicit reinforcement learning [15]. GRPO removes the separate value model and estimates the baseline from group rewards [16]. These methods show different ways to use feedback after supervised training. Our implementation follows the GRPO direction. It samples each reward group from one task. This keeps reward comparison within the same task type. This design is useful because DocVQA, ChartQA, ScienceQA, and MMMU use different answer formats and metrics.

Inference-time sampling is also important for reasoning tasks. Chain-of-thought prompting shows that models can benefit from explicit reasoning traces [17]. Self-consistency improves this idea by sampling several reasoning paths and selecting a stable answer [18]. Verifier-based methods generate multiple candidate answers and select the best one by a learned scorer [19]. Best-of-N rejection sampling has also been used with reward models in human-feedback training [20]. Our inference stage follows the broad idea of generating several candidates. However, our current validation setting selects candidates with task rewards. Therefore, it should be viewed as reward-guided Best-of-N analysis. It should not be described as a learned verifier for deployment.

3 Framework Overview

Figure 1 presents the overall training and inference pipeline. The framework contains three main parts: data preparation, model adaptation, and answer inference. The data part organizes text-rich multimodal tasks from document QA, chart QA, and external evaluation benchmarks. All samples are converted into a shared format before training. This format provides a unified interface for different task types.

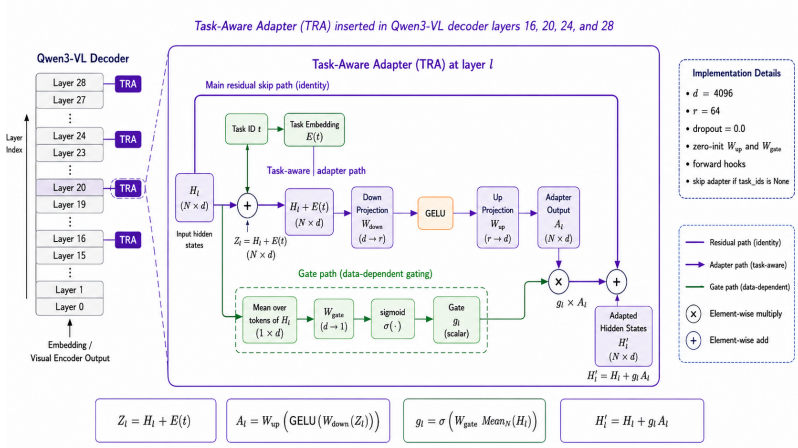

Figure 2: Structure of the proposed Task-Aware Text-Rich Adapter. The adapter adds a task embedding to the hidden states, applies a bottleneck transformation, and injects the output through a gated residual connection.

The model adaptation part starts from Qwen3-VL-8B-Instruct. The framework follows a progressive adaptation strategy. It first applies PEFT-based supervised adaptation to obtain a task-adapted model. It then introduces a Text-Rich Adapter for task-aware representation learning. Next, it uses Task-Stratified GRPO for reward-based optimization. Each stage builds on the checkpoint from the previous stage. This process links supervised learning, adapter-based modeling, and reward optimization.

The inference part uses Best-of-N generation for validation-time oracle analysis. For each input, the adapted model produces multiple candidate answers. The system then selects the final answer with reference-based task scoring in the validation setting. This setting estimates an upper bound for candidate selection, not deployment-time performance. Overall, the pipeline connects data construction, model adaptation, reward optimization, and inference. The next section describes the main method components.

4 Method

4.1 PEFT-based Supervised Adaptation

The first stage builds a task-adapted Qwen3-VL-8B-Instruct with parameter-efficient supervised tuning. Given a sample (I_i, q_i, t_i, y_i), the prompt builder uses the task id t_i to construct a structured prompt p_i. We train with answer-only supervision:

$$\mathcal{L}_{\mathrm{SFT}} = -\sum_{i=1}^{N} \sum_{j=1}^{|y_i|} \log p_\theta(y_{i,j} \mid I_i, p_i, y_{i,<j}). \tag{1}$$

Prompt and question tokens are masked, so the model learns the required answer format rather than copying the input. LoRA and DoRA are used as standard PEFT baselines, not as the main methodological novelty. Both are applied to `q_proj`, `k_proj`, `v_proj`, and `o_proj` with rank 48, alpha 96, and dropout 0.05. We compare them because text-rich tasks need new answer behavior while preserving pretrained visual-language knowledge. DoRA is selected for later stages because it improves the in-domain overall score from 0.6964 to 0.7356, especially on ChartQA.

4.2 Task-Aware Text-Rich Adapter

After PEFT adaptation, we add a lightweight Text-Rich Adapter (TRA) by forward hooks after decoder layers 16, 20, 24, and 28. TRA is designed for the part that PEFT alone does not explicitly model: different text-rich tasks require different answer forms, visual evidence, and scoring rules. Figure 2 shows the module.

Let $H_l \in \mathbb{R}^{B \times N \times d}$ be hidden states at layer l and t be the task id. TRA broadcasts a learned task embedding, applies a bottleneck residual transform, and gates its strength:

$$\begin{aligned} Z_l &= H_l + E(t), \\ A_l &= W_{\mathrm{up}}(\mathrm{GELU}(W_{\mathrm{down}} Z_l)), \\ g_l &= \sigma(W_{\mathrm{gate}} \mathrm{Mean}_N(H_l)), \qquad H'_l = H_l + g_l A_l. \end{aligned} \tag{2}$$

The implementation uses hidden size $d = 4096$ and bottleneck rank 64. The up-projection and gate are zero-initialized, so the adapter starts as an identity mapping. This reduces the risk of disrupting the DoRA checkpoint and lets training learn when task-aware residual updates are useful.

4.3 Task-Stratified GRPO

After TRA, we optimize on DocVQA and ChartQA with task-stratified GRPO. The key design is to keep each reward group task-homogeneous. At each step, the sampler chooses a task bucket, samples one input, and generates $G = 8$ answers from

Table 1: Processed datasets used in the experiments.

Dataset	Train	Validation	Usage
DocVQA	10,000	2,500	in-domain
ChartQA	15,000	1,920	in-domain
ScienceQA	–	4,242	external
MMMU	–	900	external

Table 2: Key training settings of the proposed framework.

Item	PEFT-SFT	TRA	GRPO
Init	Qwen3-VL-8B-Instruct	best SFT ckpt	best TRA ckpt
Data	DocVQA + ChartQA	DocVQA + ChartQA	DocVQA + ChartQA
Method	LoRA / DoRA	Text-Rich Adapter	task-stratified GRPO
Key params	$r = 48$, $\alpha = 96$, drop 0.05	$r = 64$, layers 16/20/24/28	group 8, temp 1.0
Optimization	lr $1e{-}4$, bs 2, accum 8	lr $2e{-}5$	lr $5e{-}6$
Training length	2 epochs	1 epoch	20k steps
Generation	max 16 tokens	max 16 tokens	max 32 tokens
Objective	answer-only loss	supervised loss	reward optimization

the current policy. DocVQA candidates receive ANLS rewards; ChartQA candidates receive normalized exact/numeric rewards. We then normalize rewards inside the group:

$$r_i^k = R_{t_i}(y_i^k, y_i^*), \qquad A_i^k = \frac{r_i^k - \mu_i}{\sigma_i + \delta}. \tag{3}$$

If the group standard deviation is nearly zero, the update is skipped. The policy term uses the mean answer-token log-probability and a clipped GRPO objective:

$$\mathcal{L}_{\mathrm{GRPO}} = -\frac{1}{G}\sum_{k=1}^{G} \min\left[\rho_i^k A_i^k, \mathrm{clip}\left(\rho_i^k, 1-\epsilon_{\mathrm{low}}, 1+\epsilon_{\mathrm{high}}\right) A_i^k\right]. \tag{4}$$

We use $\epsilon_{\mathrm{low}} = 0.2$, $\epsilon_{\mathrm{high}} = 0.28$, AdamW, and update the trainable PEFT/TRA parameters. A reference-policy KL is logged by temporarily disabling PEFT adapters, avoiding a second full model. Task stratification matters because otherwise relative advantages would compare incompatible metrics across DocVQA and ChartQA.

4.4 Validation-Time Oracle Best-of-N

After training, we generate $N \in \{1, 2, 4, 8\}$ candidate answers per validation input. For $N = 1$ the script uses greedy decoding; for $N > 1$ it samples with temperature. Reported Best-of-N results use gold-reference task rewards to select the candidate with the highest validation score. This is therefore a validation-time oracle upper bound on candidate selection. It should not be interpreted as a learned verifier or a deployable test-time scaling method. The code also contains a reference-free self-consistency fallback, but the reported curve uses oracle rewards.

5 Experiments

5.1 Experimental Setup

We evaluate the framework on four text-rich multimodal benchmarks. DocVQA and ChartQA are used for in-domain training and evaluation. ScienceQA and MMMU are used for external evaluation. All datasets are converted into the shared `UnifiedSample` format. Each sample contains an image path, a question, optional choices, a gold answer, an answer type, and task metadata. For ScienceQA, samples without images use a blank placeholder image. This keeps the multimodal input format consistent. Table 1 summarizes the processed datasets used in our experiments.

We use Qwen3-VL-8B-Instruct as the backbone model. The model is loaded with `bfloat16` precision and automatic device mapping. All supervised experiments use structured prompts. The framework is trained in three stages. The first stage applies PEFT-based supervised adaptation. The second stage adds the Text-Rich Adapter. The third stage applies task-stratified GRPO. Table 2 lists the main training settings.

Data card and preprocessing. DocVQA uses `pixparse/docvqa-wds`, ChartQA uses `HuggingFaceM4/ChartQA`, ScienceQA uses `TheMrguiller/ScienceQA`, and MMMU uses `MMMU/MMMU`; licenses and redistribution terms follow the original dataset cards. The preprocessing scripts keep examples that can be converted into `UnifiedSample`, materialize image fields when available, and do not drop ScienceQA no-image questions. Their placeholder image only satisfies the one-image model interface and adds no visual evidence, so those examples remain text-question cases. Answer normalization lowercases text, normalizes whitespace and punctuation, extracts multiple-choice labels, and compares numeric answers with a 10^{-3} absolute/relative tolerance.

Reproducibility and cost. Submission code is included under `Code/`. The main entry points are `scripts/preprocess_*.py`, `scripts/train_peft.py`, `scripts/train_grpo.py`, and `scripts/inference_best_of_n.py`; setup commands are listed in `Code/README.md`. All training configs set seed 42. The main experiments were run on a single NVIDIA RTX PRO 6000 GPU with 96GB memory. Available logs

Table 3: Main results across cumulative training stages.

Step	Setting	DocVQA	ChartQA	In-domain Overall	ScienceQA	MMMU	External Avg.
0	Qwen3-VL + prompt	0.1218	0.0760	0.1019	0.6822	0.2900	0.4861
1	LoRA SFT	0.7096	0.6792	0.6964	**0.7980**	**0.4411**	**0.6195**
2	DoRA SFT	0.7389	0.7313	0.7356	0.7888	0.4300	0.6094
3	DoRA SFT + TRA	0.8100	0.7427	0.7808	0.7893	0.4256	0.6074
4	DoRA SFT + TRA + GRPO	**0.9190**	**0.7682**	**0.8535**	0.7871	0.4156	0.6013

report DoRA SFT on 25k examples taking 1.468×10^5 seconds (about 40.8 hours), and task-stratified GRPO taking about 20.4 hours for 20k steps. The peak logged GPU memory usage is about 61.8GB.

For evaluation, DocVQA uses ANLS. ChartQA uses exact matching with numeric tolerance. ScienceQA and MMMU use multiple-choice accuracy. We report dataset-level scores and the overall average. For Best-of-N analysis, we use validation-time oracle selection: each candidate is scored against the gold answer and the highest-reward candidate is selected.

5.2 Main Results

Table 3 reports the main results across all stages. This table is a cumulative study, not a strict ablation study. Each row either adds a new training stage or uses a stronger supervised checkpoint. The in-domain overall score is computed on the combined validation set of DocVQA and ChartQA. The external average is the mean score of ScienceQA and MMMU.

The results show that supervised adaptation brings the largest early gain. The structured-prompt baseline only reaches 0.1019 on the in-domain overall score. After LoRA SFT, the score increases to 0.6964. This shows that task-specific training is needed for text-rich inputs. DoRA further improves the in-domain score to 0.7356, with a clear gain on ChartQA.

The Text-Rich Adapter improves the in-domain score from 0.7356 to 0.7808. The gain mainly comes from DocVQA, which rises from 0.7389 to 0.8100. ChartQA improves only slightly. This result suggests that the adapter is more helpful for document-style text reading and short answer extraction. However, the table still reflects cumulative training rather than an isolated adapter study.

The final GRPO stage gives the best in-domain result. The overall score reaches 0.8535. DocVQA improves strongly to 0.9190, and ChartQA also increases to 0.7682. This shows that reward-based training can improve task-level performance after supervised learning. The smaller gain on ChartQA indicates that chart reasoning remains harder in this framework.

The external results do not follow the same trend. LoRA SFT gives the best external average, with 0.6195. Later stages improve DocVQA and ChartQA, but they slightly reduce ScienceQA and MMMU. This suggests a trade-off between in-domain specialization and external generalization. Therefore, the best checkpoint should be selected according to the target task.

Ablation coverage. Table 3 supports the cumulative pipeline but does not fully isolate every component. A strict ablation should compare DoRA SFT, DoRA SFT+TRA, DoRA SFT+GRPO without TRA, the full model, and TRA variants without task embedding or without the gate. The current report covers the base, TRA, and full rows, so the GRPO-only and TRA-component effects remain limitations rather than proven independent gains.

5.3 Generalization and Error Analysis

The external results show that the later stages keep most of the model's generalization ability. LoRA SFT gives the best external average on ScienceQA and MMMU, with a score of 0.6195. The final GRPO checkpoint reaches 0.6013. The drop is 0.0182, which is relatively small. At the same time, the final checkpoint gives the best results on DocVQA and ChartQA. This result shows that the framework improves the target tasks while largely keeping external performance.

This pattern is related to the training data. TRA and GRPO are trained after supervised adaptation. Their training signals come from DocVQA and ChartQA. So the model becomes more aligned with document and chart tasks. ScienceQA and MMMU use different question styles and answer formats. Even so, the final checkpoint still keeps competitive external scores. This suggests that the later stages do not severely damage general reasoning ability.

Figure 3 gives a closer look at the error types. The correct proportion increases from 35.3% to 76.3%. Text reading failure drops from 25.4% to 4.1%. Chart reasoning failure drops from 18.6% to 4.7%. These changes match the in-domain results. The final model is better at reading text and answering chart questions. Science-figure failure is also lower than the baseline, but it remains a visible source of error.

5.4 Validation-Time Oracle Best-of-N Analysis

We further study whether additional sampled answers reveal useful headroom. For each validation input, the model generates N candidate answers with $N \in \{1, 2, 4, 8\}$. The final answer is selected by gold-reference task reward, so this setting is a validation-time oracle analysis. It is not a learned verifier and should not be read as a deployment method.

Figure 4 shows a steady oracle gain as N increases. On DocVQA, the score improves from 0.9220 at $N = 1$ to 0.9339 at $N = 8$. On ChartQA, the score improves from 0.7656 to 0.7896. The trend is clear on both datasets. More candidate

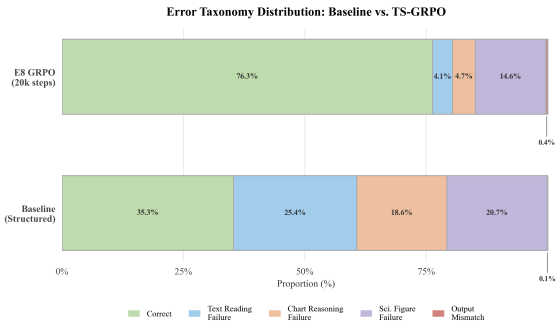

Figure 3: Automatic error taxonomy distribution for the structured baseline and the final TS-GRPO checkpoint. The final checkpoint increases the correct proportion and reduces text reading and chart reasoning failures.

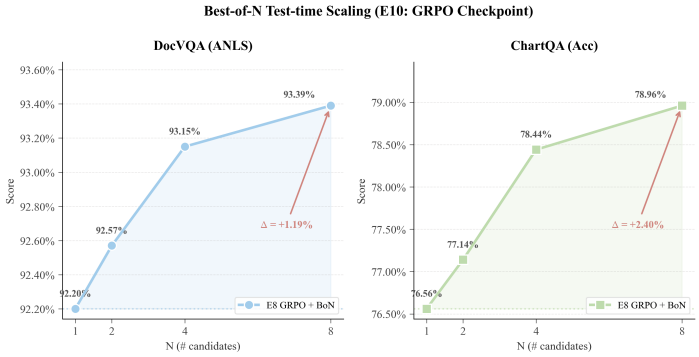

Figure 4: Validation-time oracle Best-of-N analysis on DocVQA and ChartQA. Increasing N improves validation performance when candidates are selected with gold-reference task rewards.

answers give the oracle selector a better chance to find a valid answer. The gain is larger on ChartQA, which suggests that chart questions may benefit from diverse samples if a practical verifier is later learned.

This result should be interpreted conservatively. It shows that the final model still has useful answer diversity and that candidate selection quality matters. It does not by itself prove a deployable test-time scaling method, because the reported selector uses gold answers.

6 Discussion

The staged pipeline shows that prompting alone is not enough for DocVQA and ChartQA: DoRA gives a stronger adapted checkpoint than LoRA, TRA adds task-aware hidden-state updates, and task-stratified GRPO further aligns predictions with target-task metrics. The validation-time oracle Best-of-N curve also shows useful answer diversity, but it should be interpreted only as headroom because selection uses gold-reference rewards. The main limitations are task dependence and ablation coverage. Later stages improve DocVQA and ChartQA more than ScienceQA and MMMU, Table 3 is cumulative rather than a strict ablation, and GRPO-only or TRA-component variants are still missing. Future work should add broader text-rich tasks, run the strict ablations listed above, and train a reference-free verifier before claiming a practical test-time scaling method.

7 Conclusion

This paper presents a unified framework for text-rich multimodal reasoning that combines structured data formatting, PEFT-based supervised adaptation, a task-aware Text-Rich Adapter, task-stratified GRPO, and validation-time oracle Best-of-N analysis. The final GRPO checkpoint achieves the best in-domain score while keeping competitive external generalization. The evidence supports the staged adaptation path, with the caveat that Best-of-N is oracle validation analysis and several strict ablations remain future work.

Individual Contributions. Hongrong Yang worked on data preprocessing, evaluation setup, DocVQA/ChartQA analysis, and report revision. Che Cheng worked on PEFT/TRA/GRPO implementation, training runs, checkpoint validation, and reproducibility packaging. Shaoming Li worked on Best-of-N validation analysis, figures/tables, qualitative error analysis, and final report writing.

References

[1] Minesh Mathew, Dimosthenis Karatzas, and C. V. Jawahar. Docvqa: A dataset for vqa on document images. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 2200–2209, 2021.

[2] Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. Chartqa: A benchmark for question answering about charts with visual and logical reasoning. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 2263–2279. Association for Computational Linguistics, 2022.

[3] Pan Lu, Swaroop Mishra, Tanglin Xia, Liang Qiu, Kai-Wei Chang, Song-Chun Zhu, Oyvind Tafjord, Peter Clark, and Ashwin Kalyan. Learn to explain: Multimodal reasoning via thought chains for science question answering. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

[4] Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, Cong Wei, Botao Yu, Ruibin Yuan, Renliang Sun, Ming Yin, Boyuan Zheng, Zhenzhu Yang, Yibo Liu, Wenhao Huang, Huan Sun, Yu Su, and Wenhu Chen. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[5] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

[6] Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo Molchanov, Yu-Chiang Frank Wang, Kwang-Ting Cheng, and Min-Hung Chen. Dora: Weight-decomposed low-rank adaptation. In *Proceedings of the 41st International Conference on Machine Learning*, pages 32100–32121. PMLR, 2024.

[7] Yupan Huang, Tengchao Lv, Lei Cui, Yutong Lu, and Furu Wei. Layoutlmv3: Pre-training for document ai with unified text and image masking. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 4083–4091, 2022.

[8] Geewook Kim, Teakgyu Hong, Moonbin Yim, Jeongyeon Nam, Jinyoung Park, Jinyeong Yim, Wonseok Hwang, Sangdoo Yun, Dongyoon Han, and Seunghyun Park. Ocr-free document understanding transformer. In *European Conference on Computer Vision*, pages 498–517. Springer, 2022.

[9] Fangyu Liu, Julian Eisenschlos, Francesco Piccinno, Syrine Krichene, Chenxi Pang, Kenton Lee, Mandar Joshi, Wenhu Chen, Nigel Collier, and Yasemin Altun. Deplot: One-shot visual language reasoning by plot-to-table translation. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 10381–10399. Association for Computational Linguistics, 2023.

[10] Neil Houlsby, Andrei Giurgiu, and Stanislaw Jastrzebski. Parameter-efficient transfer learning for nlp. In *Proceedings of the 36th International Conference on Machine Learning*, pages 2790–2799. PMLR, 2019.

[11] Brian Lester, Rami Al-Rfou, and Noah Constant. The power of scale for parameter-efficient prompt tuning. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3045–3059. Association for Computational Linguistics, 2021.

[12] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[13] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[14] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

[15] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

[16] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

[17] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

[18] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023.

[19] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

[20] Nisan Stiennon, Long Ouyang, Jeff Wu, Daniel M. Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul F. Christiano. Learning to summarize from human feedback. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

A Qualitative Examples

This section shows representative validation examples from the available prediction export. We include corrected cases and remaining failure cases. These examples compare the structured-prompt baseline with the trained model prediction. They show how adaptation changes answer format, text reading, and chart reasoning behavior.

Qualitative Example 1: Corrected Document Answer

Task: DocVQA
Question: When was the statement of Grocery Manufacturers of America released?
Gold Answer: July 18, 1978
Structured Baseline: July 18, 1978 — The statement of
Trained Prediction: July 18, 1978.
Note: The trained model keeps the correct date and removes extra generated text.

Qualitative Example 2: Corrected Chart Numeric Answer

Task: ChartQA
Question: What is the value of the United Kingdom?
Gold Answer: 0.354
Structured Baseline: 35.440000000000
Trained Prediction: 0.354
Note: The trained model fixes a scale error in the chart answer.

Qualitative Example 3: Corrected Chart Entity Answer

Task: ChartQA
Question: Which country has the largest bar?
Gold Answer: Nepal
Structured Baseline: Germany
Trained Prediction: Nepal
Note: The trained model selects the correct visual category from the chart.

Qualitative Example 4: Remaining Document Failure

Task: DocVQA
Question: Who has signed the letter?
Gold Answer: Leon Gutterman
Structured Baseline: Leon Gutterman
Trained Prediction: Leon Gutterman, President and Publisher.
Note: The prediction contains the correct name but over-generates extra role information.

Qualitative Example 5: Remaining Chart Reasoning Failure

Task: ChartQA
Question: What percentage of 18 to 29 year olds said they would cancel Netflix?
Gold Answer: 49
Structured Baseline: 49.000000000000
Trained Prediction: 40%
Note: The trained model gives a plausible numeric answer, but it reads the wrong chart value.